

Parametric Equations



Eliminating the parameter

- 1 Consider the curve $x = 2t + 1$, $y = t^2$.
 - a Eliminate the parameter to write y as a function of x .
 - b Identify the curve.
 - 2 Eliminate the parameter for $x = 3\cos t$, $y = 3\sin t$, $0 \leq t < 2\pi$, and describe the curve.
 - 3 For the line $x = t - 2$, $y = 2t + 3$:
 - a Find the points corresponding to $t = 0$ and $t = 2$.
 - b Eliminate the parameter to write the Cartesian equation.
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Projectile motion

- 4 A projectile follows $x = 40t$, $y = -5t^2 + 30t$ (metres, seconds).
 - a When does the projectile land ($y = 0$, $t > 0$)?
 - b Find the horizontal range.
 - c Find the maximum height.